

Validation and Official Competition Interaction Logic

Nga and Yueju, please use the following logic when designing the rules, Starter Kit, and server interaction. The important point is that **participants always run their own agent locally**. We do not run their agent code during either Validation or the Official Competition.

The Validation phase and the Official Competition should use essentially the **same observation-decision protocol**. The main difference is the time scale:

- **Validation:** one connection runs through a complete historical multi-day episode quickly.
- **Official Competition:** the same interaction happens one trading day at a time in real time.

1. Basic Participant-Server Interaction

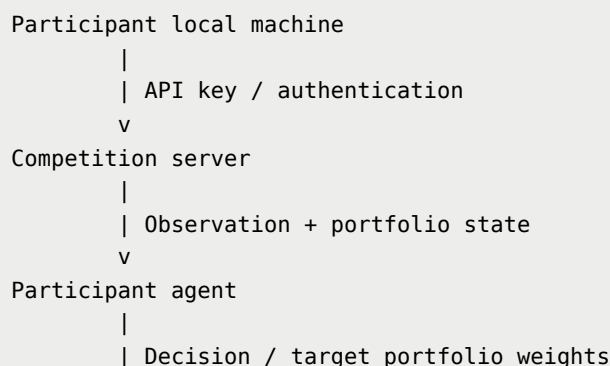
Each participant runs a local program containing:

1. their own trading agent; and
2. the official client / runner we provide in the Starter Kit.

Each team receives a unique **API key** from us.

The API key is used to identify and authenticate the team whenever the local client connects to our server.

The basic interaction is:



The participant's agent itself always stays on the participant's machine.

The server only needs to:

- authenticate the team;
- send the official observation;
- send the official current portfolio state;
- receive the participant's decision;
- validate and record the decision;
- update the portfolio according to the official execution rules.

The technical API format, endpoint names, JSON schema, etc. should be documented in the Starter Kit rather than explained in detail on the competition homepage.

2. Observation Design

The official observation should be generated after the U.S. market opens, once the current day's opening prices are available.

For trading day t , the observation should contain:

```
Observation_t
├─ market_history
│   └─ completed daily market data through trading day t-1
│       including open, high, low, close, volume
├─ current_open
│   └─ opening prices for trading day t
├─ fundamentals
│   └─ latest fundamental information available at that time
└─ portfolio_state
    ├─ current holdings / weights
    ├─ cash
    ├─ NAV
    └─ drawdown
```

The observation **must not contain the current day's full OHLCV**, because at the time the decision is made we do not yet know:

- the current day's close;
- the current day's high;
- the current day's low;
- the current day's full-day volume.

For example, on Tuesday morning:

- Monday's complete OHLCV can be included;
- Tuesday's opening price can be included;
- Tuesday's close/high/low/full-day volume cannot be included.

The `market_history` does not need to contain only the immediately previous day. It can contain the historical window or historical sequence provided by the official system, as long as all complete daily market data ends at `t-1`.

3. News

News is **not part of the organizer-provided observation**.

Participants may independently retrieve eligible open-source news while running their local agent.

For example:

- Team A receives the official observation and submits at 10:00 AM;
- Team B receives the same official observation but waits until 3:00 PM and uses additional open-source news published during the day.

That is the participant's own strategy choice.

We do not need to force everyone to submit at the same time.

The only requirement is that the information used by a participant must already be publicly available when that participant makes the corresponding decision.

4. Validation Phase

Validation is **not code submission**.

The purpose of Validation is to let participants test whether their local agent can successfully interact with our server and complete the same sequential workflow that will later be used in the Official Competition.

Each team may start up to **10 validation attempts per day**.

One validation attempt means one complete historical trading episode.

If the Official Competition contains **N** trading decision days, then each Validation attempt should also contain **N** sequential decision days.

The interaction is:

```
Participant connects to Validation server
      |
      v
Server sends historical Observation 1
      |
      v
Agent returns Decision 1
      |
      v
Server updates simulated portfolio
      |
      v
Server sends historical Observation 2
      |
      v
Agent returns Decision 2
      |
      ...
      |
      v
Server sends historical Observation N
      |
      v
Agent returns Decision N
      |
      v
Validation episode ends
      |
      v
Server returns validation metrics/results
```

The entire sequence can happen within a few seconds or minutes.

The participant does **not** need to wait one real day between validation observations.

Validation therefore tests:

- whether the API connection works;
- whether the participant can receive observations;
- whether the agent can process the observation correctly;
- whether the returned decision has the correct format;
- whether sequential portfolio state is handled correctly;
- whether the complete evaluation pipeline works.

Validation results do **not** contribute to the official final ranking.

5. Detailed Three-Day Validation Example

Assume, only for illustration, that the validation episode contains **three decision days** and only two stocks, **AAPL** and **MSFT**.

The real competition will use the full 30-stock universe.

Initial state:

```
Initial NAV: 1,000,000
Cash:       1,000,000
AAPL weight: 0
MSFT weight: 0
```

The participant starts Validation Attempt 1.

Validation Day 1

The local client connects:

```
Participant -> Server:
  API key
  Start Validation Attempt
```

The server authenticates the team and sends historical Observation 1:

```
{
  "session_date": "2025-03-03",
  "market_history": {
```

```

    "AAPL": {
      "previous_close": 240.00,
      "previous_high": 242.00,
      "previous_low": 237.00,
      "previous_volume": 51000000
    },
    "MSFT": {
      "previous_close": 410.00,
      "previous_high": 414.00,
      "previous_low": 407.00,
      "previous_volume": 22000000
    }
  },

  "current_open": {
    "AAPL": 241.00,
    "MSFT": 411.00
  },

  "fundamentals": {
    "...": "latest available fundamental information"
  },

  "portfolio_state": {
    "nav": 1000000,
    "cash": 1000000,
    "weights": {
      "AAPL": 0.0,
      "MSFT": 0.0
    },
    "drawdown": 0.0
  }
}

```

The participant's local agent processes the observation and returns:

```

{
  "target_weights": {

```

```
"AAPL": 0.10,  
"MSFT": 0.05  
}  
}
```

The server records Decision 1.

Under the next-open execution rule, this decision is executed using the historical opening prices of Validation Day 2.

Validation Day 2

Before sending Observation 2, the server:

1. executes Decision 1 at the historical Day 2 opening prices;
2. applies the official transaction fee;
3. updates holdings, cash, NAV, and portfolio weights.

The server then sends Observation 2:

```
{  
  "session_date": "2025-03-04",  
  
  "market_history": {  
    "AAPL": {  
      "previous_close": 244.00,  
      "previous_high": 246.00,  
      "previous_low": 239.00,  
      "previous_volume": 53000000  
    },  
    "MSFT": {  
      "previous_close": 409.00,  
      "previous_high": 413.00,  
      "previous_low": 405.00,  
      "previous_volume": 24000000  
    }  
  },  
  
  "current_open": {  
    "AAPL": 245.00,  

```

```

    "MSFT": 408.00
  },

  "fundamentals": {
    "...": "latest available fundamental information"
  },

  "portfolio_state": {
    "nav": 999850,
    "cash": 849850,
    "weights": {
      "AAPL": 0.10,
      "MSFT": 0.05
    },
    "drawdown": 0.00015
  }
}

```

The participant returns Decision 2:

```

{
  "target_weights": {
    "AAPL": 0.05,
    "MSFT": 0.10
  }
}

```

The server records Decision 2.

Validation Day 3

The server executes Decision 2 at the historical Day 3 open, updates the portfolio, and sends Observation 3:

```

{
  "session_date": "2025-03-05",

  "market_history": {
    "AAPL": {

```

```

        "previous_close": 248.00,
        "previous_high": 250.00,
        "previous_low": 243.00,
        "previous_volume": 55000000
    },
    "MSFT": {
        "previous_close": 412.00,
        "previous_high": 415.00,
        "previous_low": 407.00,
        "previous_volume": 23000000
    }
},

"current_open": {
    "AAPL": 247.00,
    "MSFT": 413.00
},

"fundamentals": {
    "...": "latest available fundamental information"
},

"portfolio_state": {
    "nav": 1004200,
    "cash": 849000,
    "weights": {
        "AAPL": 0.05,
        "MSFT": 0.10
    },
    "drawdown": 0.0
}
}

```

The participant returns Decision 3:

```

{
    "target_weights": {
        "AAPL": 0.08,

```

```
"MSFT": 0.07
}
}
```

The server records the decision.

If decisions are executed at the next market open, the system then uses the following historical market open as a **settlement / execution step** for Decision 3 before computing the final validation result.

No new participant decision is required during this settlement step.

The server then returns the complete validation metrics.

The full validation interaction is therefore:

```
Connect

Observation 1
->
Decision 1
->
Execute at next open

Observation 2
->
Decision 2
->
Execute at next open

Observation 3
->
Decision 3
->
Final next-open execution

Return validation results
```

This entire sequence is **one validation attempt**.

The participant can repeat the whole episode up to 10 times per day.

6. Official Competition

The Official Competition uses the same observation and decision format as Validation.

The difference is that the interaction now follows the real U.S. market calendar.

Participants do not receive the entire episode in one connection.

Instead, there is one official trading interaction per real trading day.

For each trading day:

1. the U.S. market opens;
2. the server obtains the required current opening prices;
3. the server prepares the official observation;
4. the participant connects using the team's API key;
5. the server sends that day's observation and current portfolio state;
6. the participant's local agent processes it;
7. the participant returns target portfolio weights;
8. the server records the decision if it arrives before market close;
9. the decision is executed at the next trading day's market open.

The participant can choose when to submit during the allowed market window.

However:

If the server receives the decision after the U.S. market has closed, the decision is not accepted.

Each team has only **one official trading decision per trading day**.

7. Detailed Three-Day Official Competition Example

Again, assume only **AAPL** and **MSFT** for illustration.

Initial state before the first official decision:

```
NAV:          1,000,000
Cash:         1,000,000
AAPL weight:  0
MSFT weight:  0
```

Official Day 1 - Monday

The U.S. market opens.

Once Monday's opening prices are available, the server prepares the same official Day 1 observation for all teams.

At 10:15 AM ET, Team A connects:

```
Participant -> Server:
  API key
  Request today's official session
```

The server returns:

```
{
  "session_date": "2026-10-26",

  "market_history": {
    "AAPL": {
      "previous_close": 250.00,
      "previous_high": 253.00,
      "previous_low": 247.00,
      "previous_volume": 50000000
    },
    "MSFT": {
      "previous_close": 510.00,
      "previous_high": 514.00,
      "previous_low": 507.00,
      "previous_volume": 21000000
    }
  },

  "current_open": {
    "AAPL": 251.00,
    "MSFT": 511.00
  },

  "fundamentals": {
    "...": "latest available fundamental information"
  },
```

```

"portfolio_state": {
  "nav": 1000000,
  "cash": 1000000,
  "weights": {
    "AAPL": 0.0,
    "MSFT": 0.0
  },
  "drawdown": 0.0
}
}

```

The participant may also independently retrieve eligible open-source news before making the decision.

At 1:20 PM ET, the local agent returns:

```

{
  "target_weights": {
    "AAPL": 0.10,
    "MSFT": 0.05
  }
}

```

The server receives the decision before market close and records it.

The portfolio is **not rebalanced immediately**.

Decision 1 will be executed at Tuesday's market open.

Official Day 2 - Tuesday

At Tuesday's market open:

```

Execute Monday Decision:
AAPL -> 10%
MSFT -> 5%
Cash -> remaining allocation

```

The server applies transaction costs and updates the official portfolio state.

Once Tuesday's opening prices are available, the Day 2 observation is prepared.

At 11:00 AM ET, the participant connects again.

The server returns:

```
{
  "session_date": "2026-10-27",

  "market_history": {
    "AAPL": {
      "previous_close": 254.00,
      "previous_high": 256.00,
      "previous_low": 249.00,
      "previous_volume": 52000000
    },
    "MSFT": {
      "previous_close": 508.00,
      "previous_high": 513.00,
      "previous_low": 505.00,
      "previous_volume": 23000000
    }
  },

  "current_open": {
    "AAPL": 255.00,
    "MSFT": 509.00
  },

  "fundamentals": {
    "...": "latest available fundamental information"
  },

  "portfolio_state": {
    "nav": 999850,
    "cash": 849850,
    "weights": {
      "AAPL": 0.10,
      "MSFT": 0.05
    },
    "drawdown": 0.00015
  }
}
```

```
}  
}
```

The participant's agent returns at 3:10 PM ET:

```
{  
  "target_weights": {  
    "AAPL": 0.05,  
    "MSFT": 0.10  
  }  
}
```

Because the decision arrives before market close, it is accepted.

It will be executed at Wednesday's market open.

Official Day 3 - Wednesday

At Wednesday's open:

```
Execute Tuesday Decision:  
AAPL -> 5%  
MSFT -> 10%
```

The server updates the portfolio and prepares Wednesday's observation.

The participant connects at 2:30 PM ET and receives:

```
{  
  "session_date": "2026-10-28",  
  
  "market_history": {  
    "AAPL": {  
      "previous_close": 258.00,  
      "previous_high": 260.00,  
      "previous_low": 253.00,  
      "previous_volume": 55000000  
    },  
    "MSFT": {  
      "previous_close": 506.00,
```

```

        "previous_high": 511.00,
        "previous_low": 503.00,
        "previous_volume": 25000000
    },
    {
        "current_open": {
            "AAPL": 257.00,
            "MSFT": 507.00
        },
        "fundamentals": {
            "...": "latest available fundamental information"
        },
        "portfolio_state": {
            "nav": 1003800,
            "cash": 848000,
            "weights": {
                "AAPL": 0.05,
                "MSFT": 0.10
            },
            "drawdown": 0.0
        }
    }
}

```

The participant returns:

```

{
  "target_weights": {
    "AAPL": 0.08,
    "MSFT": 0.07
  }
}

```

The server receives it at 3:55 PM ET.

Because the decision was received before market close, it is accepted.

It will be executed at Thursday's market open.

If Wednesday is the final decision day of the competition, Thursday's open must therefore be treated as the final execution / settlement point before the final portfolio value is calculated.

8. What Happens If the Decision Is Late?

Suppose the market closes at 4:00 PM ET.

The participant starts computing at 3:58 PM.

The server receives the decision at:

```
4:00:08 PM ET
```

The decision is rejected because it arrived after the official deadline.

The relevant portfolio therefore does not execute that new rebalance at the next open and remains based on the most recent valid portfolio decision, according to the official trading rules.

The deadline should be determined by the **server receive time**, not by a timestamp generated on the participant's machine.

9. Core Difference Between Validation and Official Competition

The easiest way to understand the system is:

Validation

```
One connection / attempt:

Historical Day 1
Observation -> Decision

Historical Day 2
Observation -> Decision

Historical Day 3
Observation -> Decision

...
```

```
Historical Day N
Observation -> Decision

Return validation results
```

The entire episode is completed immediately.

Each team can run up to 10 such attempts per day.

Official Competition

```
Real Trading Day 1
Observation -> Decision
WAIT until next real trading day

Real Trading Day 2
Observation -> Decision
WAIT until next real trading day

Real Trading Day 3
Observation -> Decision

...

Real Trading Day N
Observation -> Decision
```

Only one official decision is accepted per team per trading day.

10. Most Important Design Principle

Validation and Official Competition should use the same:

- API authentication logic;
- observation structure;
- portfolio-state structure;
- agent decision format;
- trading constraints;
- next-open execution logic;
- transaction-cost logic;
- evaluation engine.

The only major difference is:

Validation compresses the entire sequential episode into one historical backtest session, while the Official Competition unfolds exactly the same interaction one real trading day at a time.

This should be the core logic used consistently across the competition rules, server implementation, and Starter Kit.

11. Additional Considerations for Validation and Official Competition

1. Consider extending the duration of the Official Competition.

The current live period may be too short to provide enough trading days for meaningful evaluation. We should consider running the Official Competition for a longer period.

2. The Validation phase can be shorter than the Official Competition.

Validation does not need to reproduce the full duration of the live competition. Its main purpose is to let participants test the complete interaction pipeline and verify that their agents work correctly with our server.

3. Consider providing a public leaderboard during Validation.

After collecting the validation results from all teams each day, we can try to send the corresponding scores back to Codabench and display them as a public validation leaderboard. We still need to test whether Codabench supports this workflow conveniently. If this is difficult to implement, we can use another method to publish the validation leaderboard.

The validation leaderboard should be clearly treated as unofficial and should not affect the final competition ranking.

4. The observation and decision interfaces used in Validation must be fully aligned with those used in the Official Competition.

Participants should receive exactly the same observation structure, portfolio-state format, authentication mechanism, and decision format in both phases.

The transition from Validation to the Official Competition should not require participants to modify their code.

Ideally, the participant should be able to use exactly the same local agent and official client in both phases. The main differences should only be:

- Validation runs on historical data and completes multiple sequential trading rounds quickly;
- the Official Competition runs on real market days;
- the Official Competition may contain more trading rounds than Validation;
- Validation allows multiple attempts, while the Official Competition accepts only one official decision per team per trading day.

In other words:

```
Same participant code
|
+--> Validation Server
|   Historical observations
|   Shorter episode
|   Multiple attempts
|
+--> Official Server
|   Live observations
|   Longer episode
|   One decision per trading day
```

Participants should only need to switch from the validation environment to the official environment, rather than rewrite or adapt their agent implementation.